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Development of a prediction model for student teaching satisfaction based on 10 machine learning algorithms

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Educational data mining has become an effective tool for exploring the hidden relationships in educational data and predicting students' academic performance. Educational evaluation is an important part of the teaching process, and the traditional evaluation methods have problems such as high subjectivity and low efficiency. The advancement of machine learning technology has led to an increasing interest in data-driven methods for evaluating student teaching. This study utilizes a dataset pertaining to student evaluations from Turkey to apply and compare ten machine learning algorithms, namely Random Forest (RF), Gradient Boosting Machine (GBM), Naive Bayes (NB), K-Nearest Neighbors (KNN), Neural Networks Algorithm (Nnet), Flexible Discriminant Analysis (FDA), Support Vector Machine (SVM), Classification and Regression Trees (CART), Sparse Linear Discriminant Analysis (SLDA), and AdaBoost (ADA), in predicting student satisfaction ratings. The findings indicate that the SVM algorithm yielded the most favorable results, with performance metrics including Accuracy, Sensitivity, Specificity, Positive Predictive Value (PPV), Negative Predictive Value (NPV), Precision, Recall, F1 Score and Standard Error (SE) recorded at 0.9765, 0.9887, 0.9789, 0.9891, 0.9789, 0.9765, 0.9777, and 0.0042, respectively. Furthermore, the SVM model's predictive outcomes were elucidated by applying the SHAP (SHapley Additive exPlanations) framework. Finally, we developed the Shiny application for online prediction of learning effect satisfaction, which can provide educators with a scientific basis for evaluation and technical support for personalized teaching and curriculum optimization.

Keywords Machine learning algorithms, Student assessment, SHAP technology, Educational data mining

Educational Data Mining (EDM) has become a central driver of change in today's educational landscape. Numerous studies have shown that EDM opens up new possibilities and opportunities for technology-enhanced learning systems based on students' needs[1]. As a cross-cutting field that integrates statistics, machine learning, and data mining technologies, EDM provides key technical support for understanding complex educational environments through the in-depth exploration of educational data and extracting valuable information from raw data[2]. In this process, EDM uses machine learning and statistical analysis to uncover potential patterns behind educational data, thereby effectively improving students' knowledge and optimizing the overall effectiveness of educational institutions[3].

As a key technology in the education and teaching system, the importance of teaching evaluation runs through the whole process of education[4]. From students' perspective, teaching evaluation is not only a yardstick for assessing learning outcomes but also helps them recognize their mastery of knowledge and gaps in abilities. In turn, this enables students to adjust their learning strategies purposefully, spark their motivation, and foster their all-round development. For teachers, evaluation results are an essential basis for reflecting on teaching methods, diagnosing teaching problems, optimizing curriculum design, and helping teachers to achieve accurate teaching. In the dimension of school management, teaching evaluation provides data support for educational decision-making, which helps to grasp the overall situation of educational quality and rational allocation of educational resources.

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The research encompasses two primary processes: (1) data collection, selection of features, choice of model, preprocessing of data, training of the model, evaluation of the model, and interpretation of results. Additionally, feedback from educators will be solicited to evaluate the practicality of data-driven predictions and their influence on decision-making processes. (2) Focuses on understanding the numerous factors influencing student teaching satisfaction. It explores how factors such as course nature, course credits, teaching resources, teacher learning status, and teacher feedback timeliness influence these predictions' accuracy. The objective is to analyze and evaluate the importance of these factors in shaping the predictive model and its outcomes. This examination aims to clarify which variables are more significant in predicting teaching effectiveness satisfaction, thereby improving the model's accuracy across different student groups and educational environments.

Related literature

The traditional teaching evaluation methods have gradually revealed numerous limitations over the long term[5]. In terms of evaluation dimensions, traditional evaluations often focus on students' academic performance, neglecting their development in non-cognitive areas such as innovative abilities, practical skills, and emotional attitudes, making it challenging to reflect students' overall quality comprehensively. Another major drawback of traditional evaluations is their intense subjectivity; factors such as teachers' personal experience and emotional biases can easily interfere with the objectivity and fairness of evaluation results. Additionally, the feedback mechanism of traditional evaluation is typically lagging, with evaluation results not being promptly communicated to students and teachers, significantly reducing the evaluative impact on instructional improvements. With educational philosophy's evolution and information technology's rapid advancement, modern educational evaluation is increasingly moving toward diversification and intelligence. Cutting-edge technologies such as big data and artificial intelligence are deeply integrated into evaluation practices, significantly enhancing the precision and efficiency of evaluation.

In this context, EDM leverages its unique technological advantages to serve as an effective tool for addressing issues related to educational assessment. By analyzing multi-source educational data such as student information, educational records, exam scores, classroom participation, and question frequency, EDM can identify hidden patterns in the data, predict student academic performance, and assist educators in optimizing learning and teaching environments. Additionally, EDM can help educators predict student satisfaction with instruction, analyze internal factors influencing academic performance, and use machine learning algorithms to predict student academic performance scientifically. It is increasingly important to promote scientific and precise decision-making in education.

Various studies employ a range of machine learning classification algorithms to analyze students' academic performance across different topics.

The research topic on the prediction of students' academic performance and related methods, which focuses on predicting students' academic performance through machine learning using various types of learning data (such as course data, online behavior data, Internet usage data, etc.).

Zabriskie et al. used random forest and logistic regression models to construct early warning models for student performance in introductory calculus mechanics (Physics 1) and electromagnetism (Physics 2) courses[6]. Xing et al. constructed a prediction model based on genetic programming (GP) and used GP algorithms to process data. Compared with traditional models, this model showed significant advantages in prediction accuracy and interpretability[7].

Xu et al. extracted, calculated, and normalized feature sets such as online duration, network traffic, and connection frequency from the actual internet usage data of 4,000 college students. They then applied three machine learning algorithms—decision trees, neural networks, and support vector machines to uncover the association between internet usage behavior and academic performance and predict college students' academic performance[8].

Alshanqiti et al. combined support vector machines (SVM) and artificial neural networks (ANN), two machine learning techniques, with a teaching-based optimizer (TLBO) to develop four hybrid models incorporating anonymous information with both discrete and continuous variables based on a comprehensive learning analytics dataset. This approach aimed to improve predictions of student exam performance (failing/passing courses and final exam scores). These predictive outcomes can help students understand their academic progress, take corrective actions, and assist scholarship providers in monitoring student progress and offering support[9].

Yağcı et al. used the midterm exam scores of 1,854 undergraduate students who took Turkish Language I at a Turkish state university in the fall semester of 2019–2020 as source data to calculate and compare various machine learning algorithms, such as random forest, nearest neighbor, and support vector machine. They proposed a new model based on machine learning algorithms to predict final exam scores[10].

Alsariera et al. systematically searched online databases from 2015 to 2021, screening and evaluating 39 relevant studies to investigate existing ML methods and key characteristics for predicting student performance[11]. Luo et al. examined all blended learning data from a university in China, clustered students' online learning behaviors, and developed a new method for classifying blended courses based on the student group with the highest proportion. This method can improve the accuracy of predictions for various categories[12].

Xuan et al. used data from 253 undergraduate students in five classes participating in three courses offered by the VnCodelab interactive learning management system to train a model to predict student performance using data such as student participation patterns, practice duration, and learning activity progress tracking[13].

Dib et al. proposed an innovative approach that uses local explainability methods to analyze learners' performance. The results showed that LIME can capture the complex interactions and interrelationships among various factors. This approach enhances educators' ability to interpret model predictions, identify dropout risks, and implement more effective teaching interventions. It also improves transparency, accountability, and fairness, thereby facilitating broader and more confident adoption of these technologies by both educators and learners[14].

The research on the topic of student risk identification and intervention, which centers on identifying students at academic risk (such as students at risk of dropping out, students with learning difficulties, or high-risk groups) and proposing intervention measures.

Waheed et al. extracted handcrafted features from clickstream data in a virtual learning environment used deep artificial neural networks to predict students at risk of dropping out and proposed early intervention measures[15]. Kaur et al. collected data from a real-world high school dataset and used the open-source tool WEKA to screen potential variables. Then, they applied various classification algorithms, such as multilayer perceptrons and naive Bayes, to the data. They successfully identified slow learners through predictive data mining models[16].

Ornelas et al. developed and implemented a continuous naive Bayes classifier for the GEAR program at Rio Salado Community College, given that the previous discrete version of the prediction model had an accuracy rate of only 70% and was difficult to deploy. The new model achieved an accuracy rate of over 90% in predicting high-risk and successful students and is highly interpretable and easy to implement[17].

Adnan et al. proposed a predictive model for analyzing the problems of students with learning disabilities, assisting teachers in timely intervention, and improving student engagement and performance. The model compares and analyzes various machine learning and deep learning algorithms and constructs predictive models for different course length percentages[18].

The research on the key factors affecting academic performance, which aims to uncover core factors influencing students' or schools' academic performance (such as school characteristics, demographic attributes, curriculum features, etc.) and analyze the relationships between these factors.

Rebai et al. constructed a machine learning-based model and conducted a two-stage analysis of a sample of Tunisian secondary schools. They identified school size, competitiveness, class size, parental pressure, and the proportion of girls as key factors affecting school academic performance. The study further found that these factors have a highly nonlinear relationship with school performance, emphasizing the importance of modeling their interactions for influencing efficiency scores[19].

Fernandes et al. conducted a descriptive statistical analysis of student data from public schools in the Federal District of Brazil for the 2015 and 2016 academic years. They then created a classification model using gradient boosting machines (GBM) based on two datasets containing variables from before the start of the academic year and academic variables from two months after the beginning of the semester to predict students' academic performance at the end of the academic year. Ultimately, they found that "grades" and "absenteeism" were closely related to end-of-year performance predictions. Demographic attributes such as "community," "school," and "age" are also potential indicators of student academic success or failure[20].

Areej et al. designed a classification model based on basic student data from higher education institutions to predict student graduation performance in the early stages of degree programs. The model predicts academic performance, identifies courses that influence student performance and provides feedback to teachers and policymakers to help them create a student-centered learning environment[21].

Pachouly et al. aimed to explore the application of explainable AI (XAI) in predicting students' academic performance. By developing a transparent and accurate model, they sought to gain insights into the various factors influencing students' academic performance. They also discussed the importance of XAI in this field and demonstrated how it provides explainability and transparency[22].

Shanto et al. used a dataset containing 1205 observations to predict students' adaptability levels in online education. They employed explainable AI (XAI) techniques, such as LIME and SHAP, to identify specific features that significantly influence adaptability level predictions, among which financial status, course duration, and network type emerged as key factors. By combining robust predictive models with explainable AI, this study contributes to enhancing the effectiveness of online education and promoting students' success in the digital age[23].

The research on the construction of teaching systems and teaching effectiveness, which focuses on the development of intelligent teaching systems, the optimization of teaching evaluation methods, and the impact of teaching strategies on effectiveness, aiming to enhance the scientificity and effectiveness of the teaching process.

Kushik et al., recognizing the current integration of information technology into the educational process, assessed and predicted the impact of students' proficiency in using teaching strategies on teaching effectiveness. They conducted a preliminary experiment using the foreign language teaching process at a Russian university as an example, demonstrating their findings' expected scalability and applicability[24].

Li et al. conducted experimental performance analysis on an evaluation system based on scientific sentiment construction using NB, KNN, LR, and SVM algorithms as meta-learning algorithms for teaching evaluation texts. The results showed that the SVM meta-classifier achieved the best results with a classification accuracy rate of 92.1%[25].

Shuo et al. combined the actual needs of music teaching to construct an intelligent music teaching system based on machine learning and SVM algorithms, innovating the music teaching process. They gradually expanded the three-layer structure BPNN into a multi-layer structure, proposed a new error allocation method and the idea of calculating the error of hidden layer nodes, and trained and tested the music teaching data. They set up an experimental group and a control group to evaluate the teaching effectiveness of the system and ultimately confirmed that the system performed well[26].

Altukhi et al. proposed a framework that integrates eXplainable AI (XAI) methods into higher education applications. This framework aims to provide clear, transparent, and trustworthy AI systems. The framework we developed addresses various challenges that hinder the integration of XAI into the education sector, utilizes

Design Science Research (DSR) to ensure the rigor of the research, and examines the interaction between AI applications and educators as end-users from the perspective of affordance theory[27].

Despite the significant progress made in EDM research in areas such as predicting students' academic performance, identifying at-risk students, analyzing key influencing factors, and developing teaching systems, existing studies still have several limitations. First, current research mostly focuses on predicting objective academic outcomes (e.g., exam scores, dropout risk) while paying insufficient attention to subjective evaluation dimensions such as students' learning satisfaction and experience in the teaching process—these types of evaluations are crucial for assessing teaching effectiveness and optimizing instructional strategies. Second, although machine learning algorithms (e.g., SVM, RF, Nnet) exhibit strong predictive performance, their "black-box" nature leads to insufficient model interpretability. Few studies explicitly elaborate on the underlying mechanisms through which key factors influence outcomes, which limits the practical guiding value of these models for educational practitioners. Third, comparisons of algorithms in existing studies are often limited to three methods, lacking systematic evaluation of more mainstream algorithms, making it difficult to identify the optimal model for specific educational scenarios (e.g., evaluation data). Fourth, most studies remain at the stage of model construction and validation, failing to translate research outcomes into directly applicable tools (e.g., online prediction platforms), resulting in a disconnect between theoretical research and educational practice.

To address these shortcomings, this study focuses on the overlooked subjective evaluation dimension of predicting students' learning satisfaction. Based on a dataset of Turkish student evaluations, it systematically compares 10 mainstream machine learning algorithms and identifies SVM as the optimal model with excellent performance metrics. Furthermore, this study enhanced the model's interpretability through the SHAP framework, clarifying the key factors influencing satisfaction prediction. Finally, a Shiny application for online prediction of learning effect satisfaction was developed, which bridged the gap between research and practice, providing educators with a scientific basis for evaluation and technical support for personalized teaching and curriculum optimization.

Research methods, datasets, and evaluation metrics

Statistical methods

SHAP technology

SHAP is a model interpretation technique based on game theory that reveals the contribution of each feature to the model's prediction results, aiming to distribute the contribution of features to the prediction results fairly. Its core idea is to treat each feature as a "participant, " calculate the marginal contribution of all possible feature combinations to the prediction results, and then obtain the SHAP value of each feature through weighted averaging. This method combines global explanations (such as feature importance rankings) with local explanations (such as the basis for predicting individual samples), providing multi-dimensional model transparency.

SHAP can elucidate nonlinear relationships and interactions among features. Being model-agnostic, SHAP can be utilized with any predictive model. It employs Shapley values to articulate the contribution of each input feature to the predictions made by the model. The Shapley value, defined as the average marginal contribution of feature values across all conceivable coalitions, is computed as demonstrated in Eq. (1).

$$\phi i(f,x) = \sum\nolimits_{z' \subseteq x'} \frac{|z'|!(M-|z'|-1)!}{M!} \left[f_x(z') - f_x(z' \backslash \{i\})\right] \tag{1}$$

$\phi i(f,x)$ represents the Shapley value contribution of feature *i* to the prediction result of model *f* for a specific input *x*. z' is a subset of the input feature set x', which x' generally consists of features other than target feature *i*. *M* represents the total number of features.

$f_x(z')$ refers to the model's predicted values under a feature subset z', where the feature subset z' is selected from the original input *x*. $f_x(z' \backslash \{i\})$ is the predicted value of the model under a subset that does not include feature i. $\frac{|z'|!(M-|z'|-1)!}{M!}$ is the weight of subset z', which reflects the proportion of importance of this subset among all possible combinations.$f_x(z') - f_x(z' \backslash \{i\})$ is the marginal contribution to the predicted value when the x feature is added to subset z'.

K-means algorithm

K-means is an unsupervised learning algorithm that aims to divide n data points into K mutually exclusive clusters so that data points within each cluster are as close as possible (high intra-cluster similarity) and clusters are as dispersed as possible (low inter-cluster similarity). Its core idea is to minimize the within-cluster sum of squares (WCSS) through iterative optimization, which is the sum of the squares of the distances between data points and their respective cluster centers.

Let the dataset is$X = \{x1, x2, ..., xn\}$, each data point $xi \in \mathbb{R}^d$(d is the feature dimension), and the cluster center set be $C = \{c1, c2, ..., cK\}$, where $c_k \in \mathbb{R}\ominus^d$ is the center of the *k* cluster.

The distance between data points $xi$and cluster centers c_k is usually measured using Euclidean distance, which is calculated using the formula in Eq. (2).

$$d(x_i, c_k) = \sqrt{\sum\nolimits_{j=1}^{d} (x_{ij} - c_{kj})^2} \tag{2}$$

Among them, x_{ij} is the *j* feature value of the data pointxi, and $c_{kj}$is the *j* feature value of the cluster centerc_k. The algorithm iterates repeatedly, assigning each data point $xi$to the cluster corresponding to the nearest cluster center $c_k$and updating the cluster centers until a stopping condition is met, such as the change in cluster centers being less than a certain threshold or reaching a preset number of iterations.

Boruta algorithm
The Boruta algorithm is a feature selection algorithm designed to select the most important features for the prediction target from high-dimensional data sets. Its core idea is to screen features by comparing the importance of the original features with randomly generated "shadow features" in the random forest model.

Assuming we have T decision trees, the formula $I(X)$for calculating the importance of feature X can be expressed as Eq. (3):

$$I(X) = \frac{1}{T}\sum\nolimits_{t=1}^{T} \Delta G_t(X) \tag{3}$$

Among them, $\Delta G_t(X)$is the reduction in the Gini index caused by splitting using feature X in the t decision tree.

Suppose there are n importance values $I_1, I_2, I_3, \ldots, I_n$ for the original features and m importance values $J_1, J_2, J_3, \ldots, J_m$ for the shadow features. We need to test whether the mean importance value of the original featuresμ_I is greater than the mean importance value of the shadow featuresμ_J. The formula for the t-test is given by Eq. (4):

$$t = \frac{I - J}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \tag{4}$$

Among them, $I = \frac{1}{n}\sum\nolimits_{i=1}^{n} I_i$, $J = \frac{1}{m}\sum\nolimits_{i=1}^{m} J_j$, S_p are the combined standard deviations, the calculation formula is given by Eq. (5).

$$S_p = \sqrt{\frac{(n-1)s_I^2 + (m-1)s_J^2}{n+m-2}} \tag{5}$$

wheres_I^2ands_J^2 are the sample variances of the importance values of the original features and shadow features, respectively. The calculated t-value is compared with the critical value at a given significance level. If t is greater than the critical value, the importance of the original feature is considered to be significantly higher than that of the shadow feature.

The algorithm determines important features through multiple iterations. This process continues until all features are classified as important or unimportant, or until the preset number of iterations is reached.

Dataset and feature selection
The dataset used in this paper is sourced from the Turkish Student Evaluation Project[28]. It includes multidimensional features such as students' classroom participation, homework completion, and exam scores. The dataset contains 5,820 records, each comprising 33 feature variables.

The dataset itself is of good quality with no missing values. To investigate the impact of each metric on student evaluation outcomes, we applied the K-means clustering algorithm to the dataset. Figure 1 shows the elbow plot and silhouette coefficient plot of the dataset. Based on the optimization results, the dataset is divided into three categories. The distribution plot of the number of each category is presented in Fig. 2, with the counts of each category being 1232, 2230, and 2358 respectively. The number distribution across the categories is relatively balanced.

Using the Boruta algorithm, we selected 30 essential features from the 34 fields, including features Q10, and Q23, for subsequent machine learning. After applying Boruta, we found that feature Q10 consistently showed higher importance scores than its shadow features across multiple runs. Compared to other features, feature Q23 made a greater contribution to reducing prediction errors in the model. The other selected features also

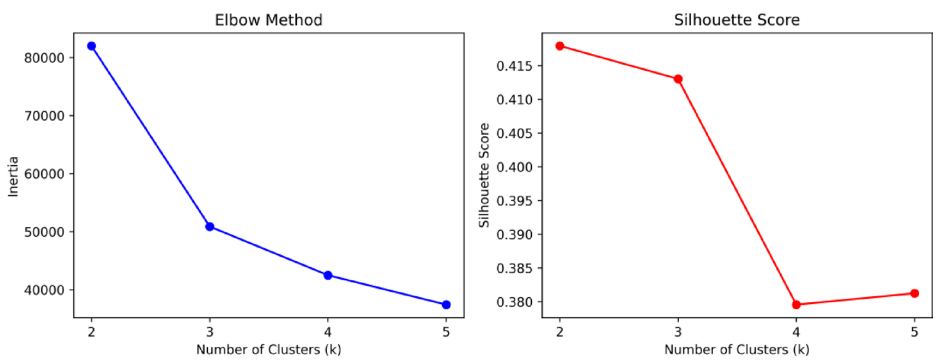

Fig. 1. Elbow diagram and contour coefficient diagram.

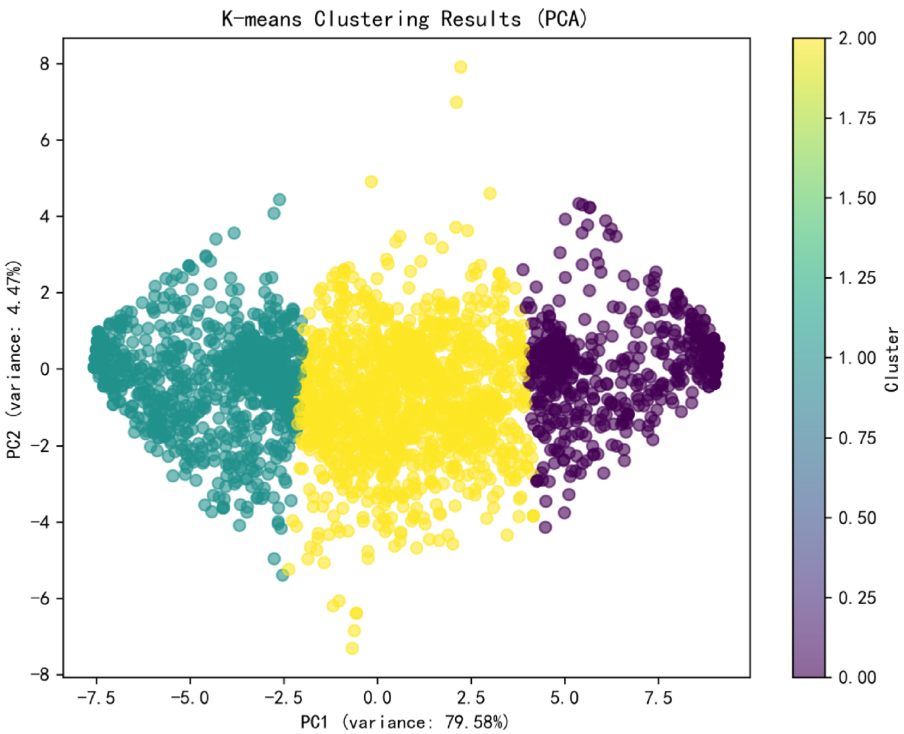

Fig. 2. K-means clustering diagram.

conformed to the screening rules of the Boruta algorithm. Figure 3 shows the weight ranking information of the 30 features selected by the Boruta algorithm, where a higher ranking indicates a greater influence of the weight.

Technical approach

For this academic performance prediction task, we use a dataset containing 5820 sample. The dataset is divided into training and testing sets using the random function sample, with 70% allocated to the training set and 30% to the testing set, which can strike a balance between "model learning sufficiency" and "generalization evaluation reliability". Meanwhile, the subsequent use of 5-fold cross-validation for further model optimization is also compatible with this 70 – 30 split.

The study employs 10 machine learning algorithms, such as RF, GBM, NB, KNN, Nnet, FDA, SVM, CART, SLDA, and ADA, to predict and evaluate student learning outcomes. Each model undergoes 5-fold cross-validation or parameter optimization (For the random forest algorithm, the hyperparameters in the grid search were configured as mtry is (2, 5, 10), splitrule is gini, min.node.size is (1, 5, 10). For the SVM algorithm, the optimal cost and gamma parameters are searched through 5-fold cross-validation. The candidate values for the cost parameter are (0.1, 1, 10, 100, 1000), and the candidate values for the gamma parameter are (0.001, 0.01, 0.1, 1, 10).

We evaluate the model's performance on the test set using machine learning evaluation metrics such as Accuracy (95% CI), Sensitivity, Specificity, PPV, NPV, Precision, Recall, and F1.

All analyses in this paper primarily use R software (4.3.0) and Python software (3.11.0). We utilized packages such as plotROC, caret, pROC, and e1071 in R software to plot ROC curves, and used Python software to plot SHAP plots for parameter visualization. Finally, we employed the shiny package to develop an online learning effectiveness satisfaction prediction app.

Figure 4 presents the technical workflow of this study. The dataset is partitioned. 10 machine learning algorithms are adopted for comparative analysis. Finally, feature visualization is performed using SHAP and Shiny.

Evaluation metrics

To assess the efficacy of the prediction model, we employ standard metrics frequently utilized in student grade prediction frameworks, specifically accuracy, precision, recall, and F1 score[29].

True Positive (TP) refers to the count of genuinely positive instances that the model accurately identifies as positive.

False Positive (FP) denotes the number of instances that are actually negative but are incorrectly classified as positive by the model.

True Negative (TN) indicates the number of genuinely negative instances that are correctly identified as negative by the model.

False Negative (FN) represents the count of positive instances that are misclassified as negative by the model.

Precision is shown in Eq. (6):

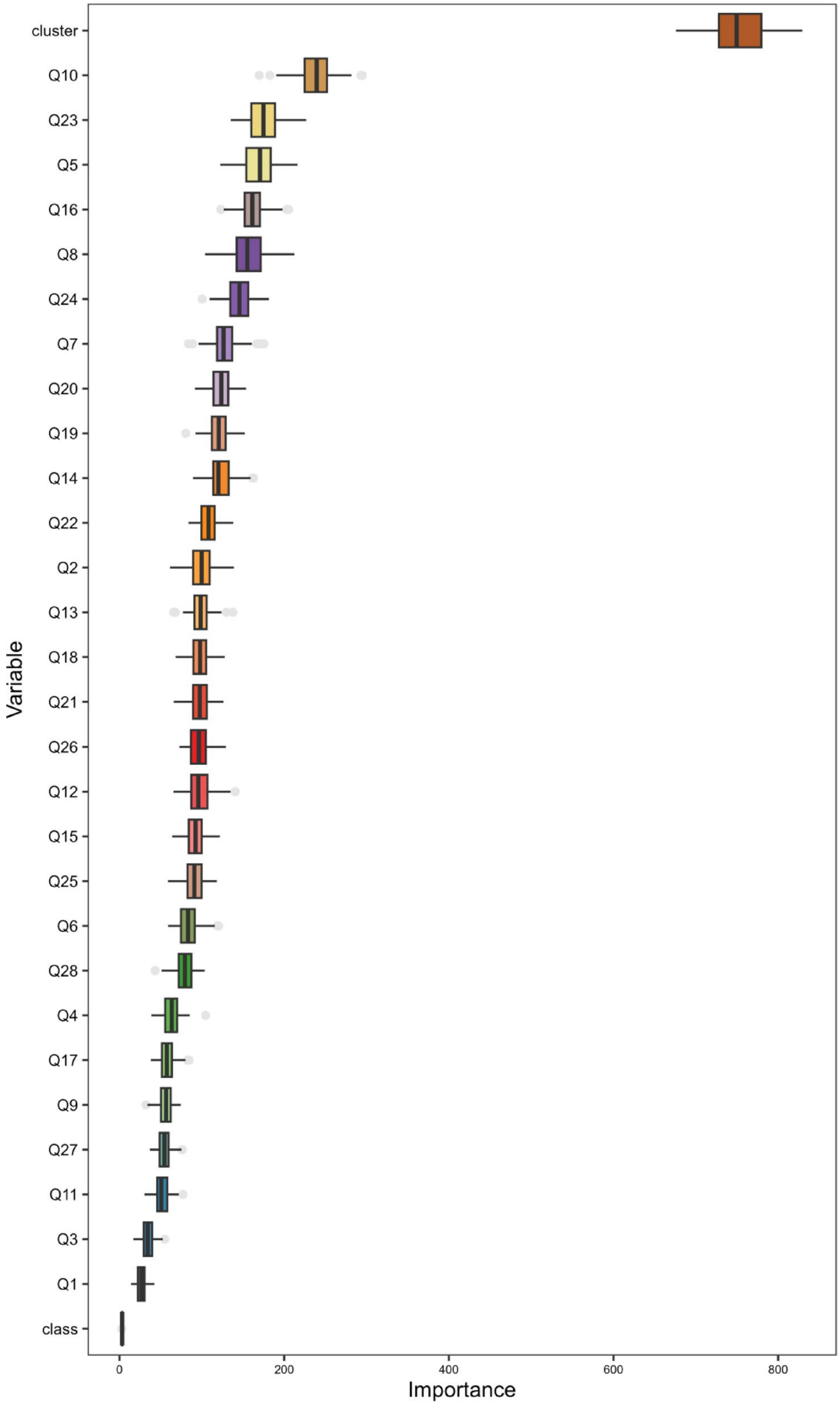

Fig. 3. Evaluation indicators selected based on the Boruta algorithm.

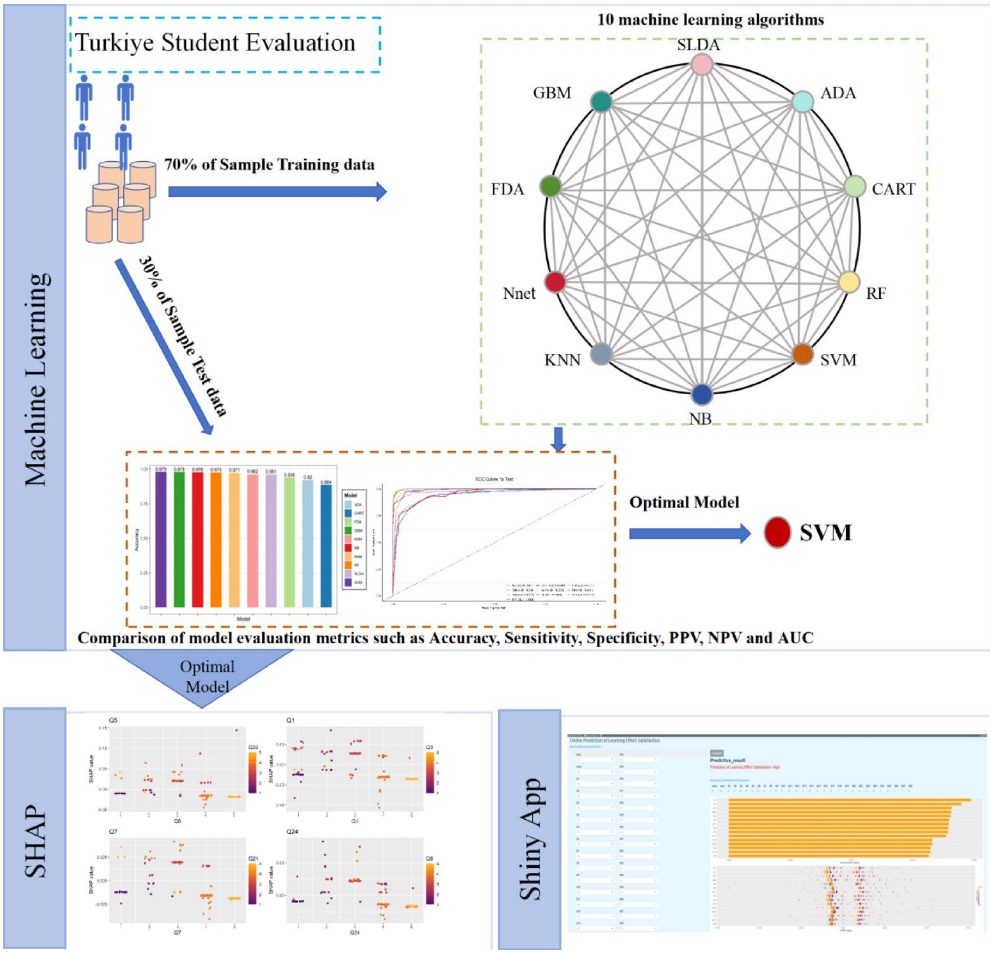

Fig. 4. Research methodology diagram.

$$Precision = \frac{TP}{TP + FP} \tag{6}$$

The proportion of truly positive samples among those predicted to be positive reflects the accuracy of the model's positive predictions.

The recall rate formula is shown in Eq. (7):

$$Recall = \frac{TP}{TP + FN} \tag{7}$$

The proportion of samples that are actually positive examples and are correctly predicted as positive examples by the model measures the model's ability to capture positive examples.

Specificity is shown in Eq. (8):

$$Specificity = \frac{TN}{TN + FP} \tag{8}$$

The proportion of samples that are actually negative examples and are correctly predicted as negative examples by the model reflects the model's ability to identify negative examples.

The accuracy formula is shown in Eq. (9):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{9}$$

The proportion of samples correctly predicted by the model out of the total number of samples reflects the model's overall predictive accuracy.

The F1 score formula is shown in Eq. (10):

Model	Accuracy (95%CI)	Sensitivity	Specificity	PPV	NPV	Precision	Recall	F1	SE
RF	0.9748 (0.9663, 0.9816)	0.9680	0.9859	0.9796	0.9877	0.9796	0.9680	0.9734	0.0064
GBM	0.9777 (0.9696, 0.9841)	0.9765	0.9881	0.9775	0.9883	0.9775	0.9765	0.9770	0.0063
NB	0.9760 (0.9676, 0.9826)	0.9789	0.9874	0.9757	0.9870	0.9757	0.9789	0.9771	0.0041
KNN	0.9616 (0.9515, 0.9702)	0.9640	0.9803	0.9581	0.9795	0.9581	0.9640	0.9608	0.0067
Nnet	0.9708 (0.9618, 0.9782)	0.9689	0.9846	0.9702	0.9848	0.9702	0.9689	0.9695	0.0058
FDA	0.9365 (0.924, 0.9474)	0.9384	0.9671	0.9329	0.9666	0.9329	0.9384	0.9351	0.1435
SVM	**0.9788 (0.9709, 0.985)**	**0.9765**	**0.9887**	**0.9789**	**0.9891**	**0.9789**	**0.9765**	**0.9777**	0.0042
CART	0.8838 (0.8678, 0.8985)	0.8838	0.9404	0.8739	0.9395	0.8739	0.8838	0.8781	0.1833
SLDA	0.9605 (0.9503, 0.9691)	0.9643	0.9797	0.9580	0.9790	0.9580	0.9643	0.9605	0.0068
ADA	0.9199 (0.9061, 0.9322)	0.9077	0.9570	0.9217	0.9593	0.9217	0.9077	0.9138	0.0993

Table 1. Results of 10 machine learning models.

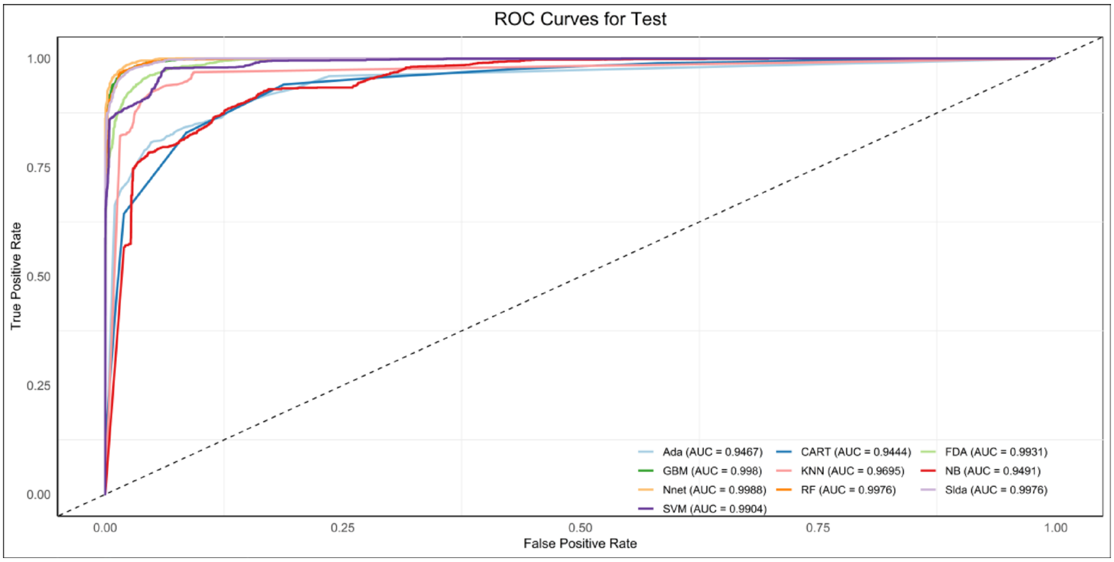

Fig. 5. ROC curves for 10 machine learning models.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{10}$$

F1 combines precision and recall metrics to balance their importance, with values ranging from 0 to 1. The closer the value is to 1, the better the model performance.

Results and analysis

Comparison and analysis of machine learning model performance

We implemented ten machine learning models on the training dataset, with the performance metrics for each model presented in Table 1. The area under the curve (AUC) values for the ten models ranged from 0.9444 to 0.9988, and the receiver operating characteristic (ROC) curves for all models.

As can be seen from Figs. 5 and 6, we found that the SVM model had the highest accuracy (0.9788, CI: 0.9709–0.985), the CART model had the lowest Accuracy (0.8838, CI: 0.8678–0.8985), while the GBM, RF, and NB models also achieved very good Accuracy scores of 0.9777, 0.974, and 0.9760, respectively.

To assess the efficacy of the ten machine learning models, we performed a comparative analysis utilizing seven distinct metrics: Sensitivity, Specificity, Positive Predictive Value (PPV), Negative Predictive Value (NPV), Precision, Recall, F1 Score and Standard Error (SE). The SVM model demonstrated the best performance among the 10 machine learning models, with metric values of 0.9765, 0.9887, 0.9789, 0.9891, 0.9789, 0.9765, 0.9777, and 0.0042 respectively. Models such as GBM, RF, and NB also performed very well, with metric values closely matching those of the SVM model. However, the CART model performed poorly on the test set, with relatively low metric values.

After conducting a comprehensive analysis of the predictive performance of different models, we found that the SVM model has the highest accuracy and robustness. Meanwhile, we employed the McNemar statistical test to separately compare the SVM algorithm with the other 9 machine learning algorithms. Table 2 presents the results of these statistical tests, where a p-value < 0.05 indicates a significant difference between the models.

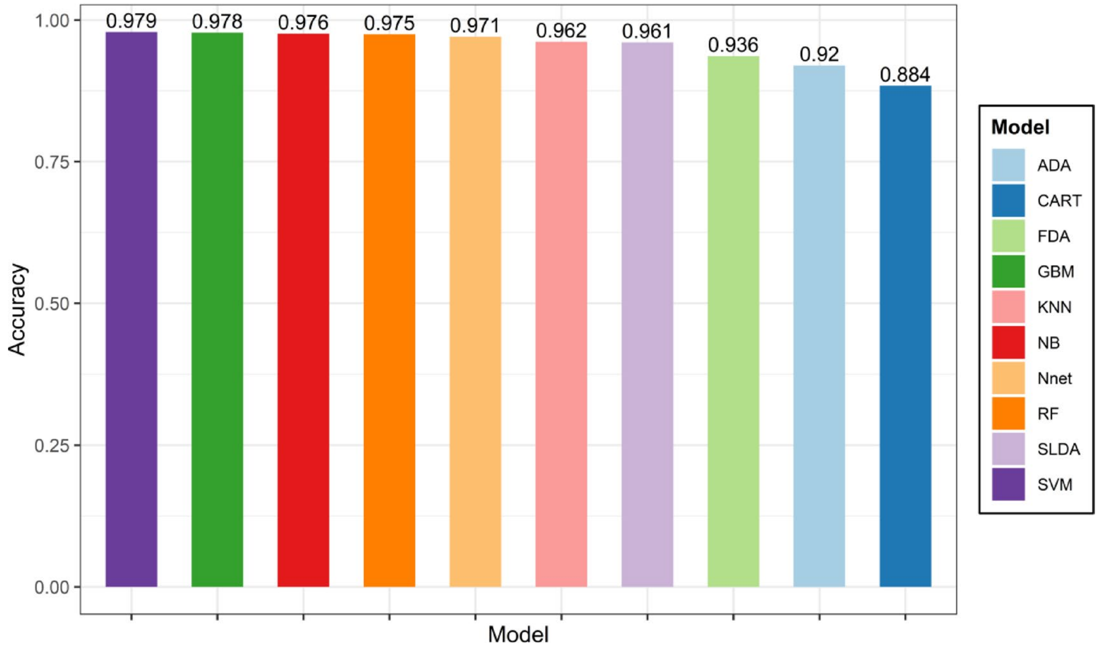

Fig. 6. Accuracy of 10 machine learning models.

Model	P_Value
SVM vs. RF	8.803958e−01
SVM vs. NB	1.000000e+00
SVM vs. KNN	2.227532e−03
SVM vs. Nnet	7.835388e−02
SVM vs. FDA	4.946256e−12
SVM vs. GBM	4.582583e−01
SVM vs. CART	1.548311e−34
SVM vs. SLDA	1.550317e−03
SVM vs. ADA	9.936464e−19

Table 2. The McNemar statistical test of SVM.

Therefore, we selected the SVM model for subsequent SHAP analysis and developed an online application for Shiny APP.

Model explanation analysis

To further explore how various features in the support vector machine (SVM) model affect learning outcomes, this study used SHAP technology to perform an explainability analysis of the model. Figures 7 and 8 respectively show the weighted influence of different features on the model's performance results.

Through SHAP analysis, we identified 30 indicators, including Q5, Q1, Q7, Q24, Q8, Q6, and Q11, all of which influence learning effectiveness satisfaction. Q5, Q1, and Q7 are ranked among the top three with a significant advantage, emerging as the key indicators influencing model output effectiveness. Among these, positive changes in the Q5 indicator show a strong correlation with improvements in learning effectiveness satisfaction, with each unit change yielding significant positive feedback; the Q1 indicator plays a stabilizing regulatory role in model decision-making, and fluctuations in its values directly affect the overall trend of model predictions; while the Q7 indicator exhibits a nonlinear relationship with learning effectiveness satisfaction, with even minor changes within specific numerical ranges potentially triggering significant fluctuations in satisfaction levels. The identification of these key indicators not only reveals the logic behind the SVM model's decision-making process and provides clear directional guidance for optimizing learning strategies and adjusting model parameters in subsequent stages.

To further understand the correlation between key indicators, we analyzed the correlation between the four indicators Q5, Q1, Q7, and Q24. Figure 9 shows the correlations among these four different features.

Different values of Q5 correspond to scatter points of various colors, representing different values of the Q22 indicator. The scatter points are relatively dispersed, indicating no strong linear correlation between Q5 and Q22.

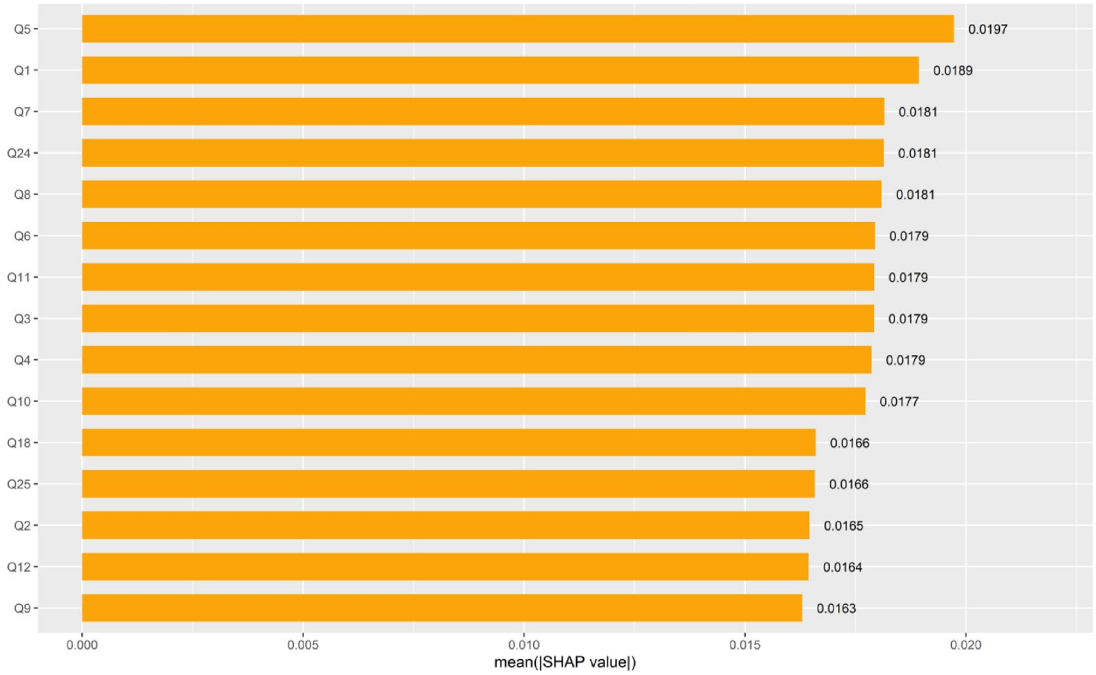

Fig. 7. Ranking chart of the importance of each feature in the model.

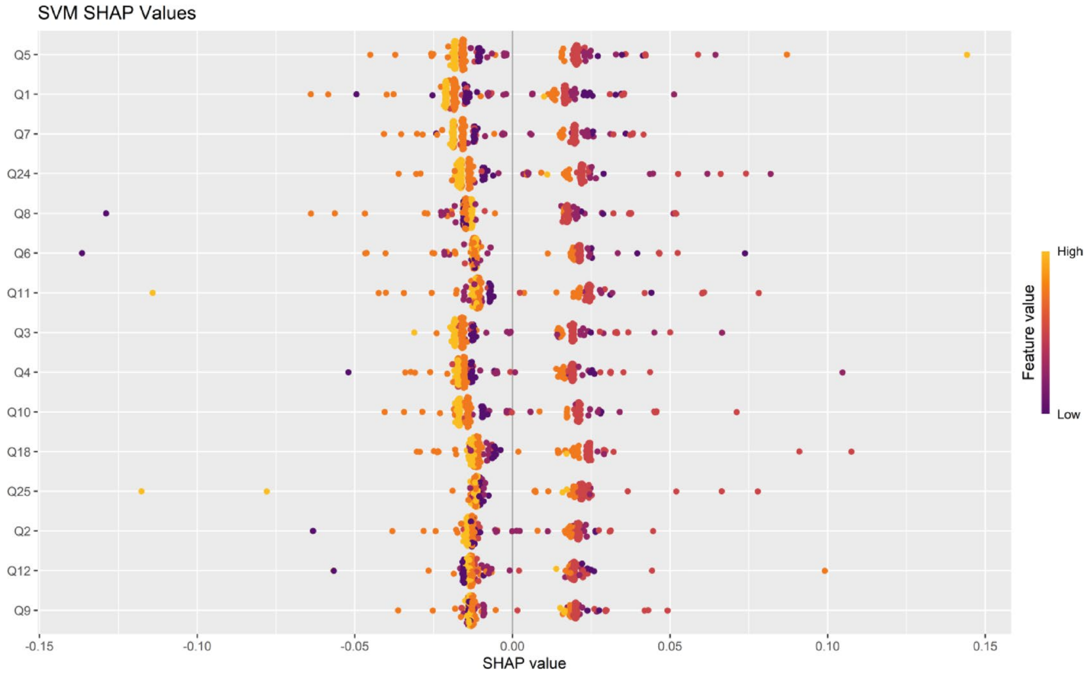

Fig. 8. SHAP feature explanation diagram.

Under specific Q5 values, the range of SHAP values fluctuates significantly, reflecting notable differences in the influence of features.

With Q1 values as the x-axis and SHAP values as the y-axis, the colors correspond to the Q3 indicator. The scatter plot shows no evident pattern, indicating that Q1 and Q3 have a low correlation and Q1's influence on the model output is unstable.

The scatter plot is scattered, indicating that Q7 and Q21 have a weak correlation, and Q7's influence on the model output varies significantly under different values.

The scatter plot shows no significant clustering trend, indicating that Q24 and Q8 have a weak correlation, and the influence of Q24 on the model output varies significantly under different conditions.

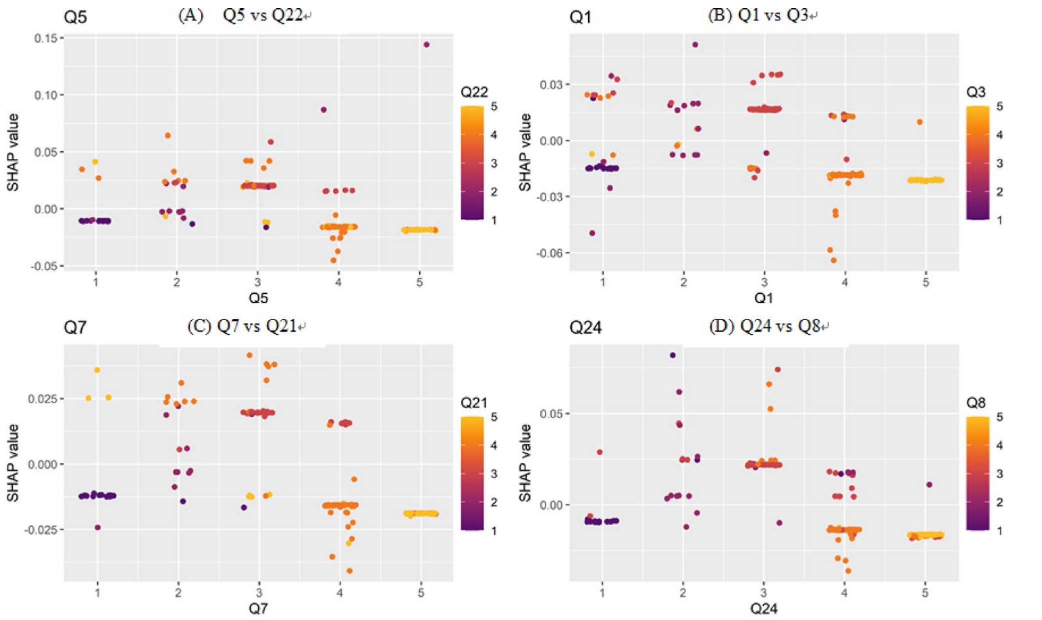

Fig. 9. SHAP feature correlation map.

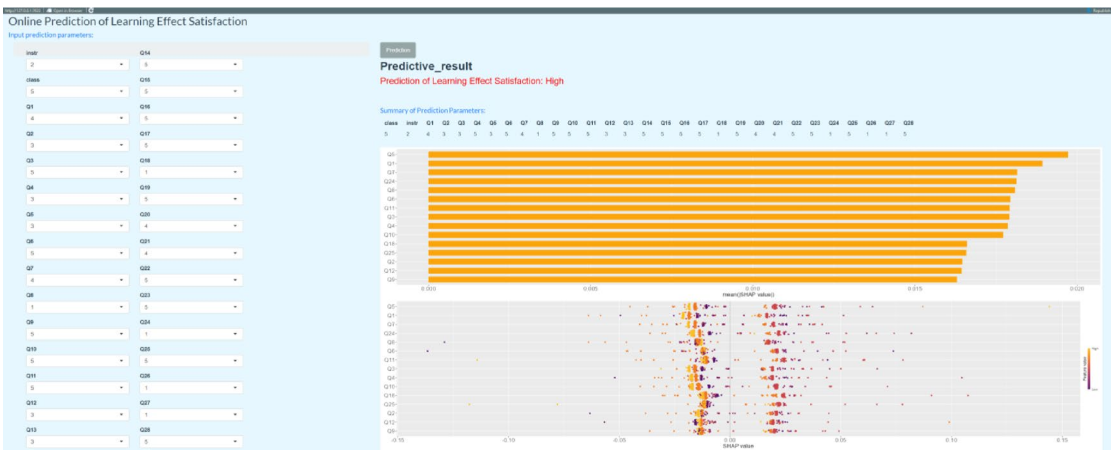

Fig. 10. Online prediction results for learning satisfaction.

Overall, the correlations between these four indicators and the indicators represented by their corresponding colors are insignificant, and the influence of features on the model output is complex and unstable.

Online prediction of learning effectiveness satisfaction using the shinny website

To achieve visual interaction between model prediction and feature interpretation, we developed an online prediction system based on the Shiny framework for learning effectiveness satisfaction. It can be accessed at the following URL: https://medicalpredictor.shinyapps.io/Prediction_of_Learning_Effect/.

Figure 10 shows the online prediction system developed in this study. The left side of the system is the parameter input interface, while the right side displays the model's judgment on the input parameters and real-time shows the weighted influence of each indicator on the evaluation result.

To demonstrate the prediction system, the system's prediction parameters have been set to include 30 indicators, such as Q5, Q1, Q7, Q24, Q8, Q6, and Q11. Users can input prediction parameters to obtain real-time learning satisfaction prediction results. This system not only lowers the barrier to using machine learning technology but also supports dynamic teaching interventions—for example, for students predicted to have "low satisfaction," the system promptly assesses the course teaching status. It appropriately adjusts and improves the curriculum and teaching methods.

Discussion

The outstanding performance of SVM is attributed to its unique ensemble learning mechanism and robust feature interaction capabilities, enabling it to handle complex nonlinear relationships in educational evaluation data easily. From the perspective of ensemble learning, SVM divides data by finding the optimal hyperplane, which not only maximizes the class margin but also effectively integrates information from multiple dimensions, similar to combining weak classifiers in ensemble learning. In educational evaluation scenarios, students' learning outcomes are influenced by various factors such as classroom participation, homework quality, exam scores, and learning attitudes, which intertwine to form complex data relationships. SVM can comprehensively consider these factors by optimizing the hyperplane, thereby accurately evaluating and predicting students' learning performance.

Regarding feature interaction capabilities, SVM leverages kernel functions to map raw data into high-dimensional spaces, enabling linearly inseparable data in low-dimensional spaces to become linearly separable in high-dimensional spaces. This effectively captures non-linear feature interactions within the data. Take classroom participation and homework quality in educational evaluation as examples; these factors are not linearly related and may exhibit synergistic effects. For instance, active classroom participation can deepen students' understanding of knowledge, thereby improving homework quality; conversely, high-quality homework feedback can encourage students to engage in more targeted learning in the classroom, further enhancing classroom participation. SVM uses kernel functions to transform such complex synergistic effects into linear relationships in high-dimensional space, enabling accurate identification and utilization of these feature interaction insights to achieve more precise assessments of learning outcomes.

This combination of ensemble learning mechanisms and feature interaction capabilities enables SVM to effectively analyze the complex nonlinear relationships in educational evaluation data, demonstrating outstanding performance and providing a scientific and reliable analytical tool for educational evaluation.

Educational evaluation data typically exhibit complex nonlinear relationships and contain numerous interrelated features. The CART model makes decisions based on a tree structure, primarily by recursively partitioning the feature space to construct the tree. It performs well on linearly separable data but may struggle to accurately describe and fit complex nonlinear relationships due to the limitations of its tree structure. For example, it may fail to adequately consider the combined effects of multiple features such as classroom participation, assignment quality, and students' learning attitudes, thereby impacting the model's predictive performance.

Through SHAP analysis, this study identified multiple indicators that influence learning satisfaction. The diversity of course categories (e.g., the ratio of theoretical to practical courses) directly impacts students' interest alignment and knowledge acquisition. SHAP values indicate that interdisciplinary course design can significantly enhance satisfaction. The appropriateness of course difficulty reflects the "challenge-skill balance" theory, where excessive difficulty leads to anxiety, while insufficient difficulty results in a lack of motivation. The timeliness and effectiveness of Q&A support enhance students' self-efficacy by reducing feelings of learning barriers, with this effect being particularly prominent in online education settings.

Additionally, the contribution of indicators such as classroom time utilization, teaching affinity, and relevance to career development suggests that optimizing teaching process design (e.g., reducing ineffective interactions), strengthening emotional connections between teachers and students (e.g., regular feedback communication), and clarifying the alignment between course objectives and career goals are potential leverage points for improving satisfaction.

To improve learning satisfaction in a targeted manner, we can dynamically adjust course difficulty and progress based on real-time learning data (such as classroom interaction frequency and assignment completion rate), use SVM models to predict students with low satisfaction, and intervene in their learning in advance. Based on SHAP feature importance weights, we can prioritize the development of a knowledge base for frequently asked questions while integrating AI chatbots to provide 24/7 real-time Q&A support, reducing waiting times. For the "teaching affinity" metric, we can design training modules for teachers on communication skills and emotional management and continuously optimize teaching behaviors through anonymous student feedback.

Although the SVM model demonstrates superior performance, we must remain vigilant about its "black box" characteristics, which may weaken the interpretability of educational outcomes. Future research can be conducted from the following three aspects to build a more comprehensive teaching feedback system: (1) Combine the predictive capabilities of SVM with the causal inference advantages of Bayesian networks to develop a hybrid model framework tailored for educational scenarios; (2) Integrate unstructured data such as classroom videos and speech emotion recognition to construct a fine-grained satisfaction prediction system; (3) Validate the timeliness of indicator impacts through long-term data tracking to identify the dynamic evolution patterns of factors driving learning effectiveness satisfaction.

In summary, this study provides quantitative evidence for precise educational quality interventions through machine learning model comparisons and interpretability analysis, while also proposing practical pathways for methodological innovation in educational data science.

Conclusion

This paper explores the application value of machine learning technology in teaching evaluation. A total of 10 machine learning algorithms, including RF, GBM, NB, KNN, Nnet, FDA, SVM, CART, SLDA, and ADA, were used to predict and compare students' evaluation satisfaction results. The results showed that the SVM model had the highest accuracy and robustness. SHAP technology quantified the contribution of each link in the teaching process, revealing the core role of indicators such as course category and course difficulty, thereby providing data-driven decision-making support for optimizing teaching strategies. To achieve visual interaction between model prediction and feature interpretation, we developed an online prediction system based on the Shiny framework for learning effectiveness satisfaction. This study integrates machine learning with explainable technology, providing methodological and practical tool support for the intelligent transformation of educational evaluation.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Declarations

Competing interests
The authors declare no competing interests.

Additional information
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